

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 16)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1°C temperature difference (U-value).

Table 1

Glazing	Heat loss / $\text{W m}^{-2} \text{ K}^{-1}$
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

0 1

. 1

A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12°C . Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer

W

0 1

. 2

Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 1

. 3

Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 1 . 4

Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

0 1 . 5

State **one** reason why insulation thickness has an optimum value.

[1 mark]

10

0 2

Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg ⁻¹
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

0 2 . 1

Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

02 . 2

Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer

tonnes

02 . 3

Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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02 . 4

Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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02 . 5

Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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10

03 . 1

Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

03 . 2

Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

03 . 3

Describe how human activities have altered the phosphorus cycle.

[3 marks]

03 . 4

Explain **two** strategies for the sustainable management of phosphorus supplies.

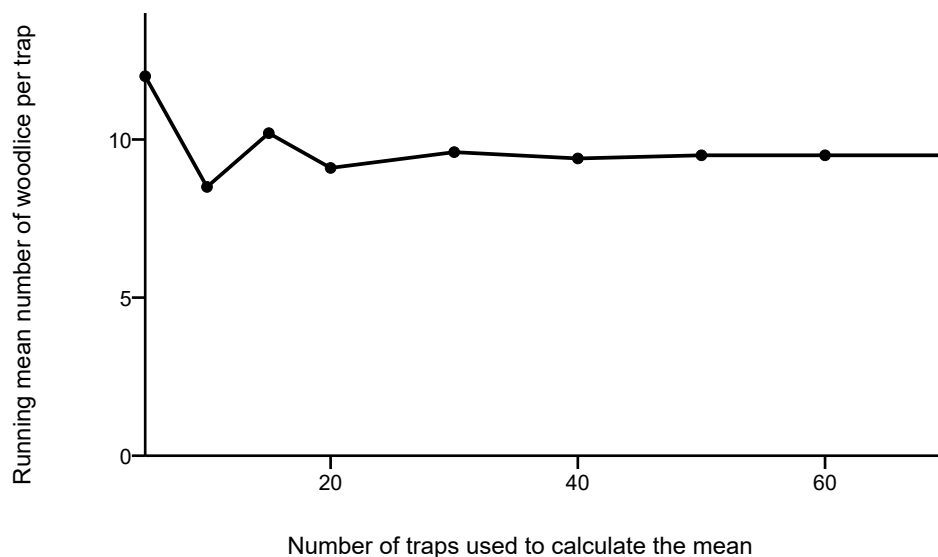
[2 marks]

10

0 4

Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 4

. 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 4

. 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 4

. 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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04 . 4

Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

10

0 5

Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1**0 5****1**

Describe the trend shown in Figure 1.

[2 marks]**0 5****2**

Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer

%

0 5**3**

DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

0 5 . 4

Suggest why the recovery after 1970 was slow.

[2 marks]

10

0 6 . 1

Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

0 6 . 2

Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

06 . 3

Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

06 . 4

Explain what is meant by **synergism** and give an example.

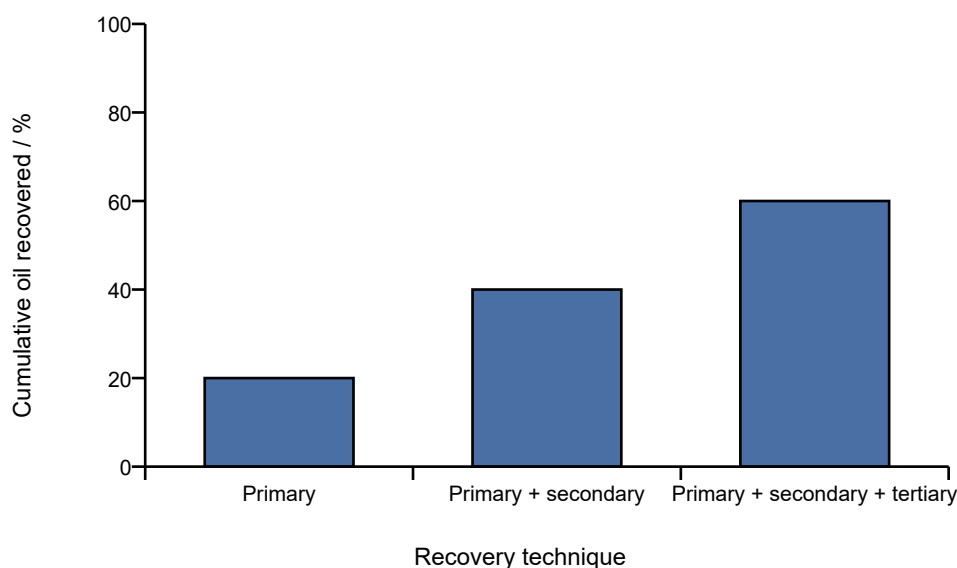
[3 marks]

10

07

Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



07

1

Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

07

2

Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

07

3

Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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2

3

0 7 . 4 Explain **three** advantages of directional drilling.

[3 marks]

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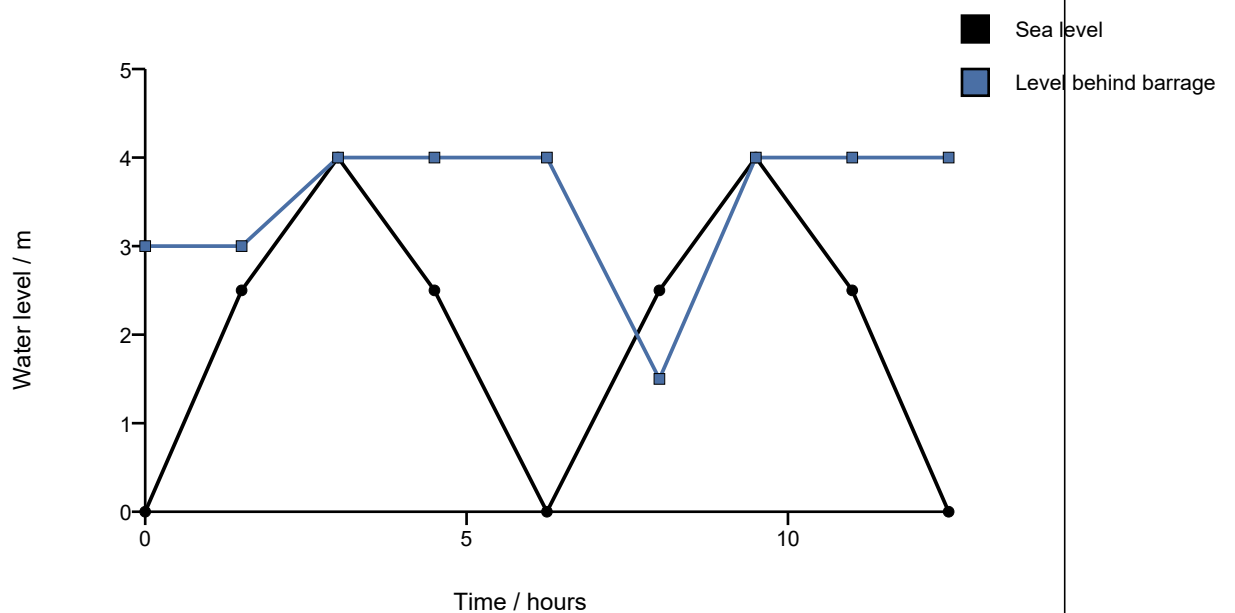
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10

0 8 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 8 . 1

Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

0 8 . 2

Explain why tidal power is intermittent but predictable.

[2 marks]

0 8 . 3

Explain the difference between spring and neap tides and why this affects output.

[2 marks]

0 8 . 4

Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

0 8 . 5Give **two** coastal features that make a site suitable for tidal power.**[2 marks]**

1

2

10

0 9 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 9 . **1** Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

0 9 . **2** Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

0 9 . **3** Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

5

10.1

Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

Extra space

10 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

OR

1 1 . 2

Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS